COMSATS University Islamabad
Department of Management Sciences · BS (Business Data Analytics)

Configuration Control Dashboard
for any configurable product

One live view of every part, version and approved change

Student photo

STUDENT
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Supervisor photo

SUPERVISOR
[Supervisor Name]
Management Sciences

Project Goals

1. Track every item and variant
2. Approve every change
3. Detect drift automatically
4. Audit-ready in seconds
5. Work for any product type

Works for Any Product

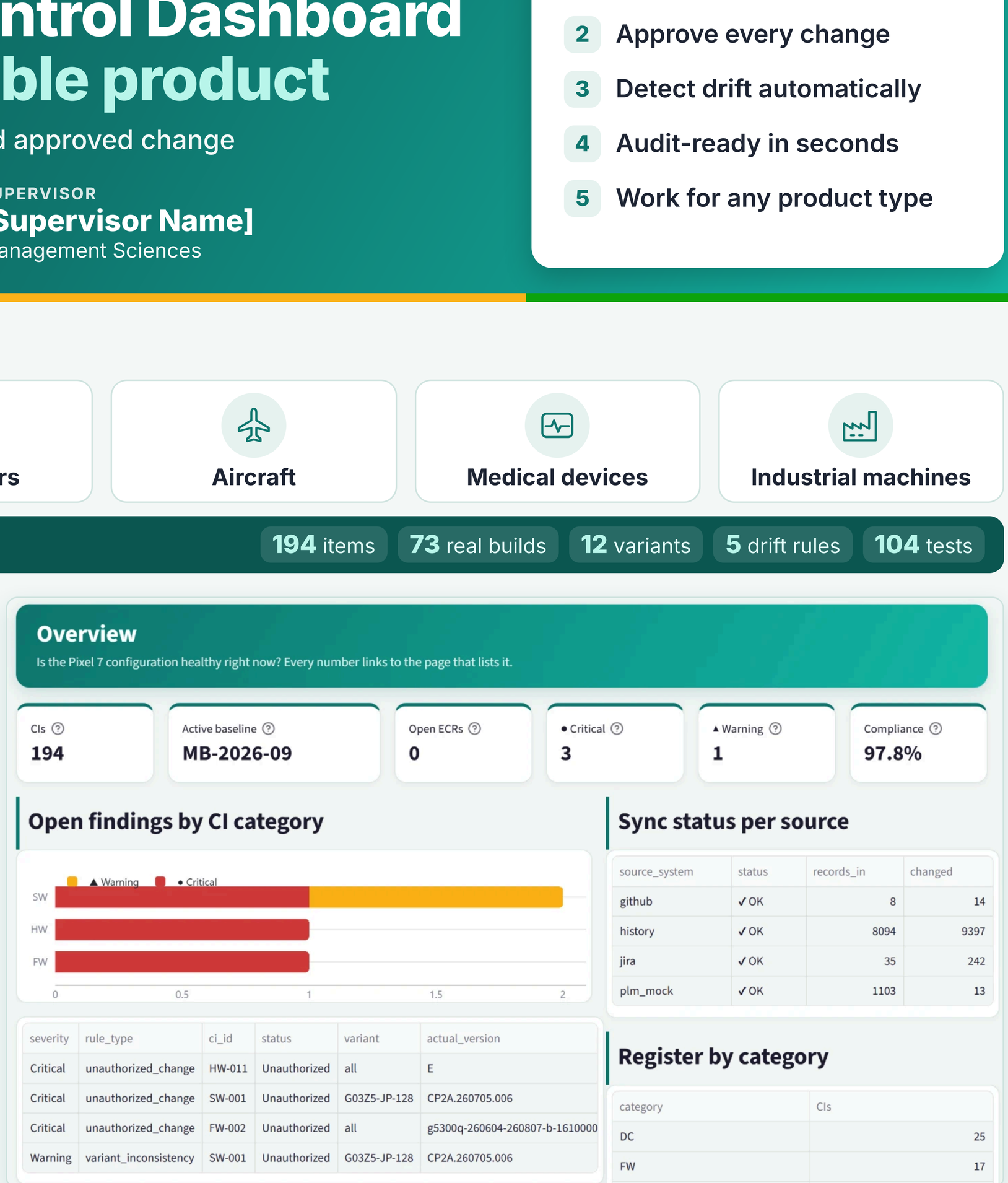
- Smartphones
- Electric cars
- Aircraft
- Medical devices
- Industrial machines

Proven on a real case study: **Google Pixel 7** — **194** items · **73** real builds · **12** variants · **5** drift rules · **104** tests

What is the CCD?

A web dashboard that shows **what is inside every product variant** and raises an alarm when anything **changes without approval**.

- Register of all parts
- Approved baselines
- Change requests (Jira)
- Drift alerts
- Variant matrix
- One-click audit report



Case study (Pixel 7): unapproved firmware change caught instantly **3 Critical** **1 Warning**

How It Works

APPROVED baseline + **ACTUAL** deployed state + **AUTHORIZED** change requests

✓ Match: compliant

● Mismatch: alarm

Software repo (GitHub) · Change requests (Jira) · PLM / BOMs · Release data

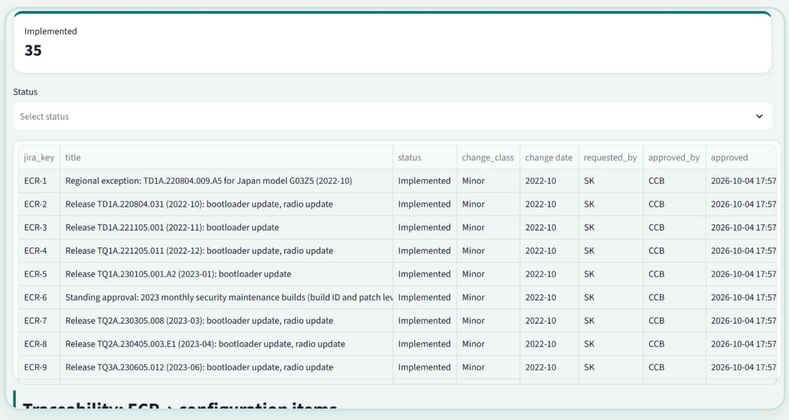
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Sync → Database → Engine → Dashboard

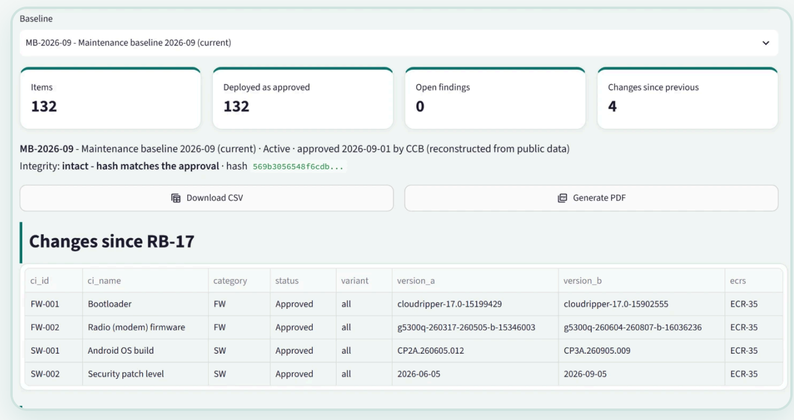
Inside the Dashboard



Variant Matrix red = differs



Change Requests traced to approval



Audit Report one-click PDF

Project Plan

1. Research
2. Data & Sync
3. Engine
4. Dashboard
5. Testing

✓ All phases complete

Deliverables

- ✓ 7-page dashboard
- ✓ Live data sync
- ✓ Drift detection
- ✓ Audit reports

Built With

Python · Streamlit · FastAPI · SQLAlchemy · Plotly · GitHub API · Jira API